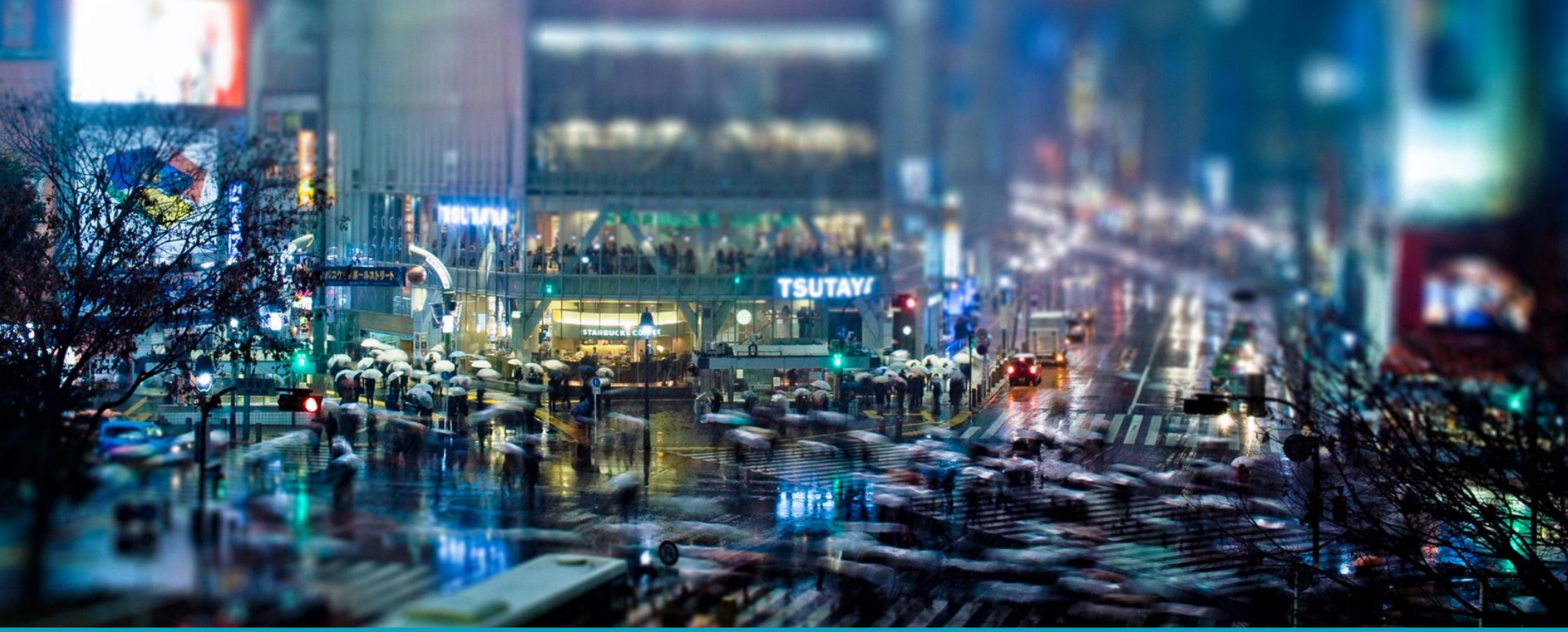


Transforming Data into Action

Berner-Architekten-Treffen

Patrick Bönzli
Manuel Linder

Inhalt

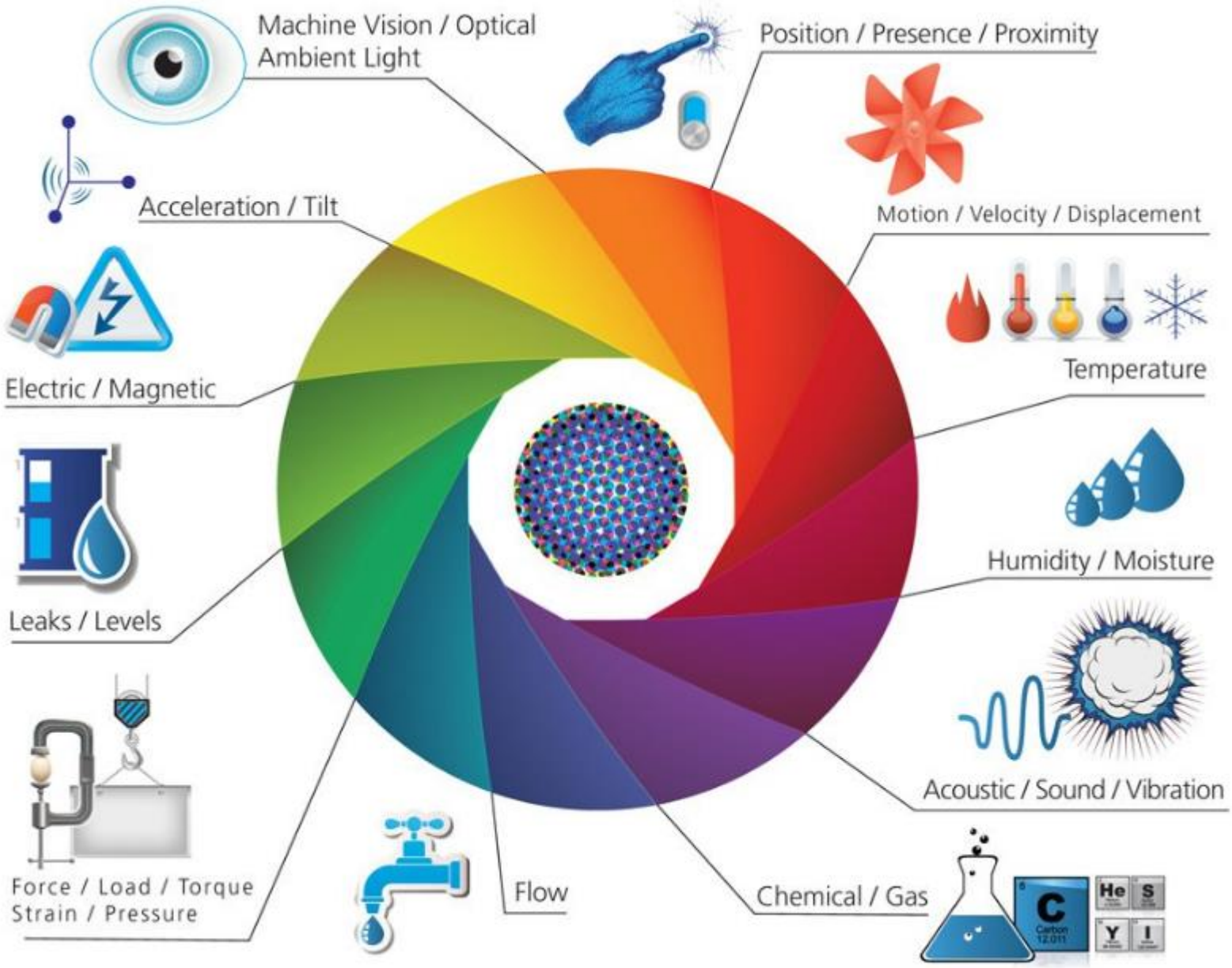
Vision und Impulse

Technologie und Wissen

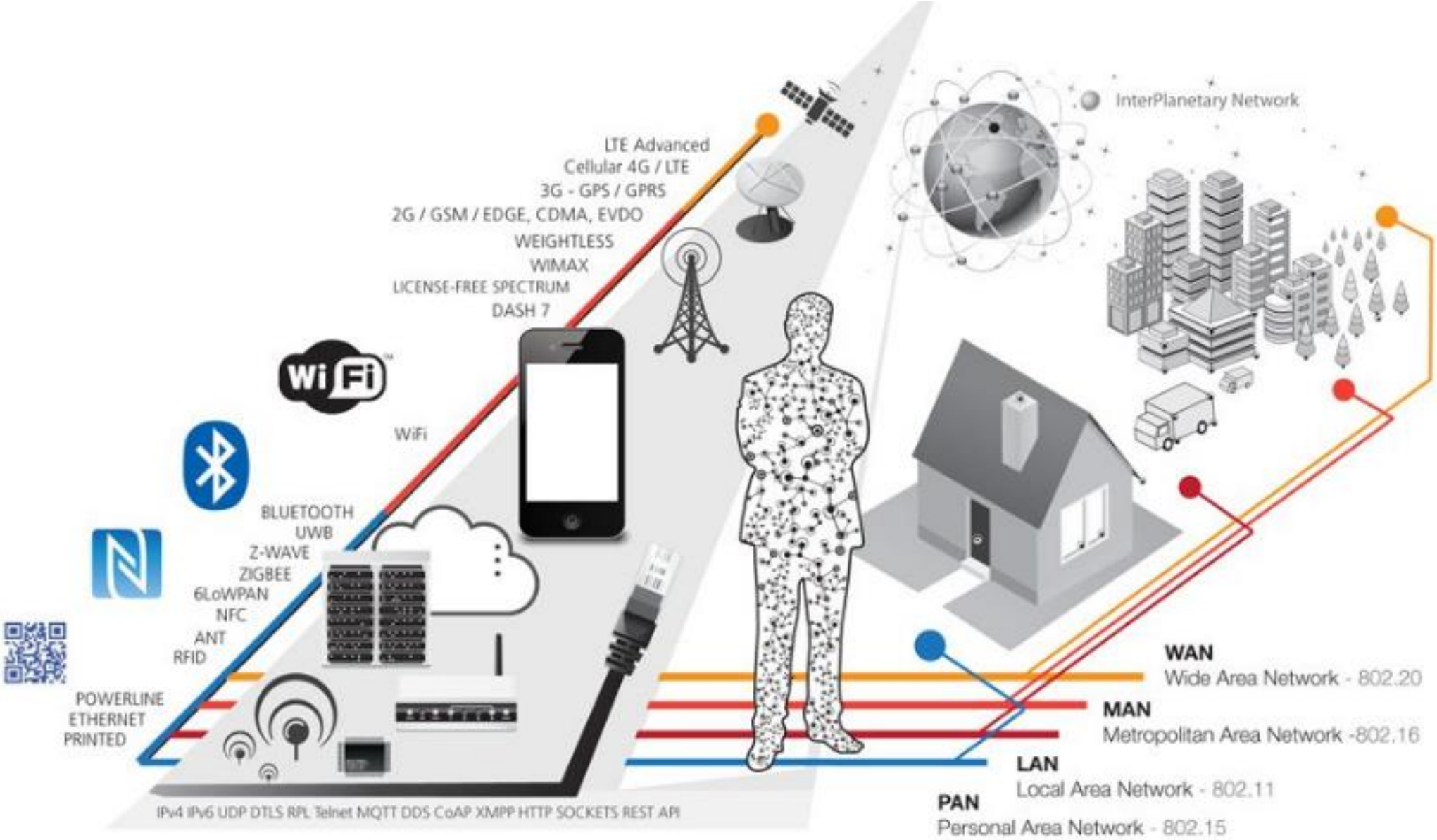
Anwendung und Erfahrungen

IMPULSE

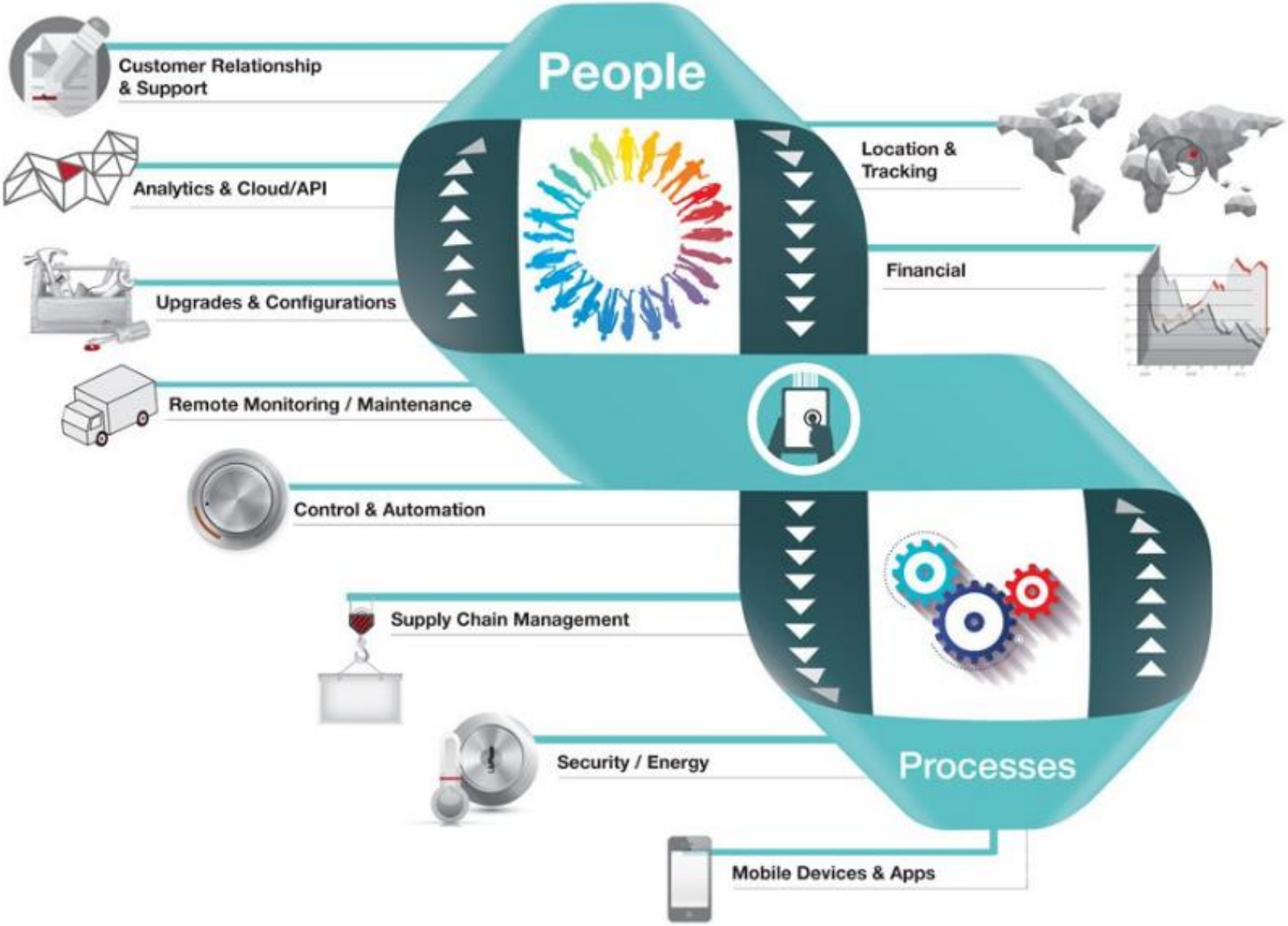
Sensoren


Machine Vision / Optical
Ambient Light
Position / Presence / Proximity
Acceleration / Tilt
Motion / Velocity / Displacement
Electric / Magnetic
Temperature
Leaks / Levels
Humidity / Moisture
Acoustic / Sound / Vibration
Force / Load / Torque
Strain / Pressure
Flow
Chemical / Gas

Connectivity



People & Processes

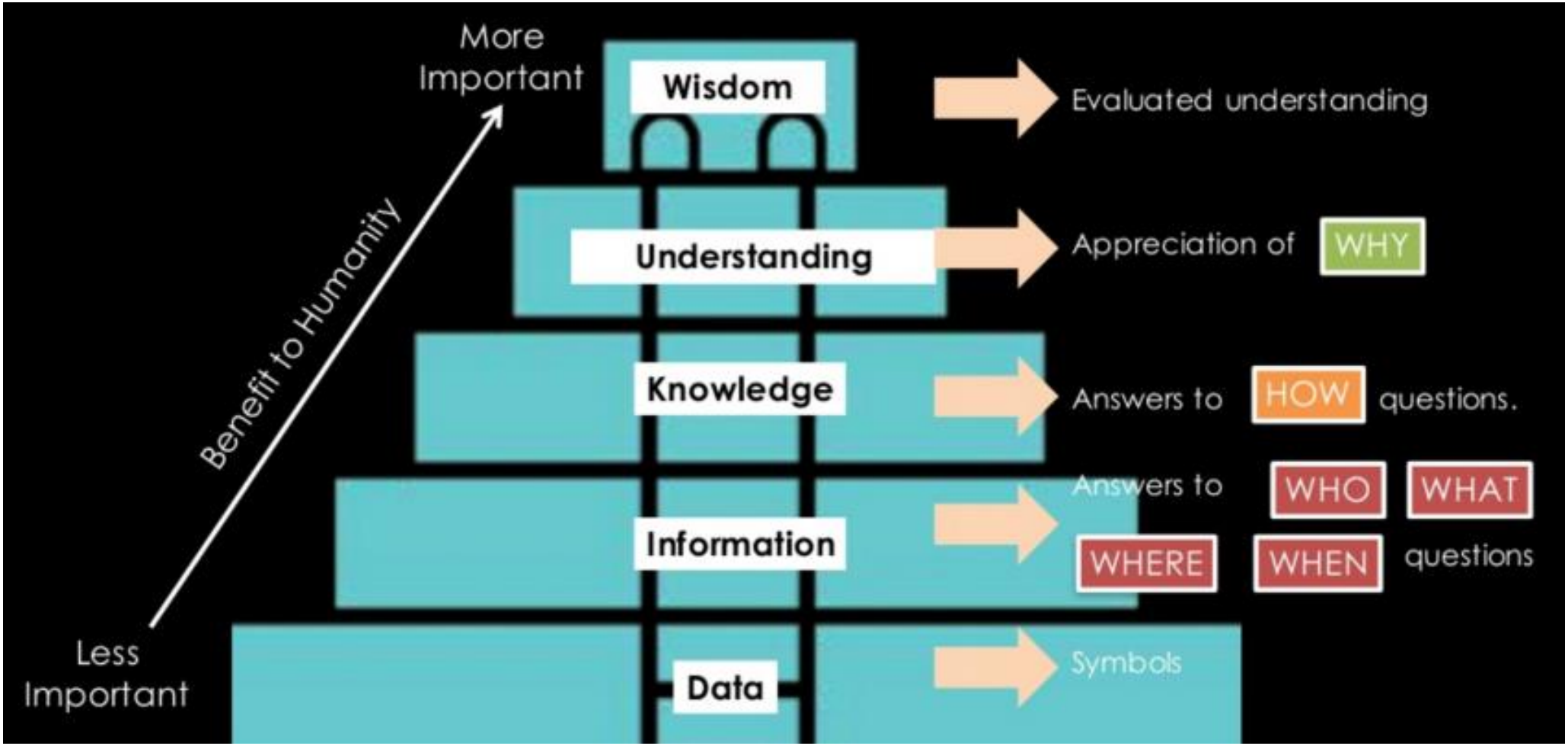


Blend the Data

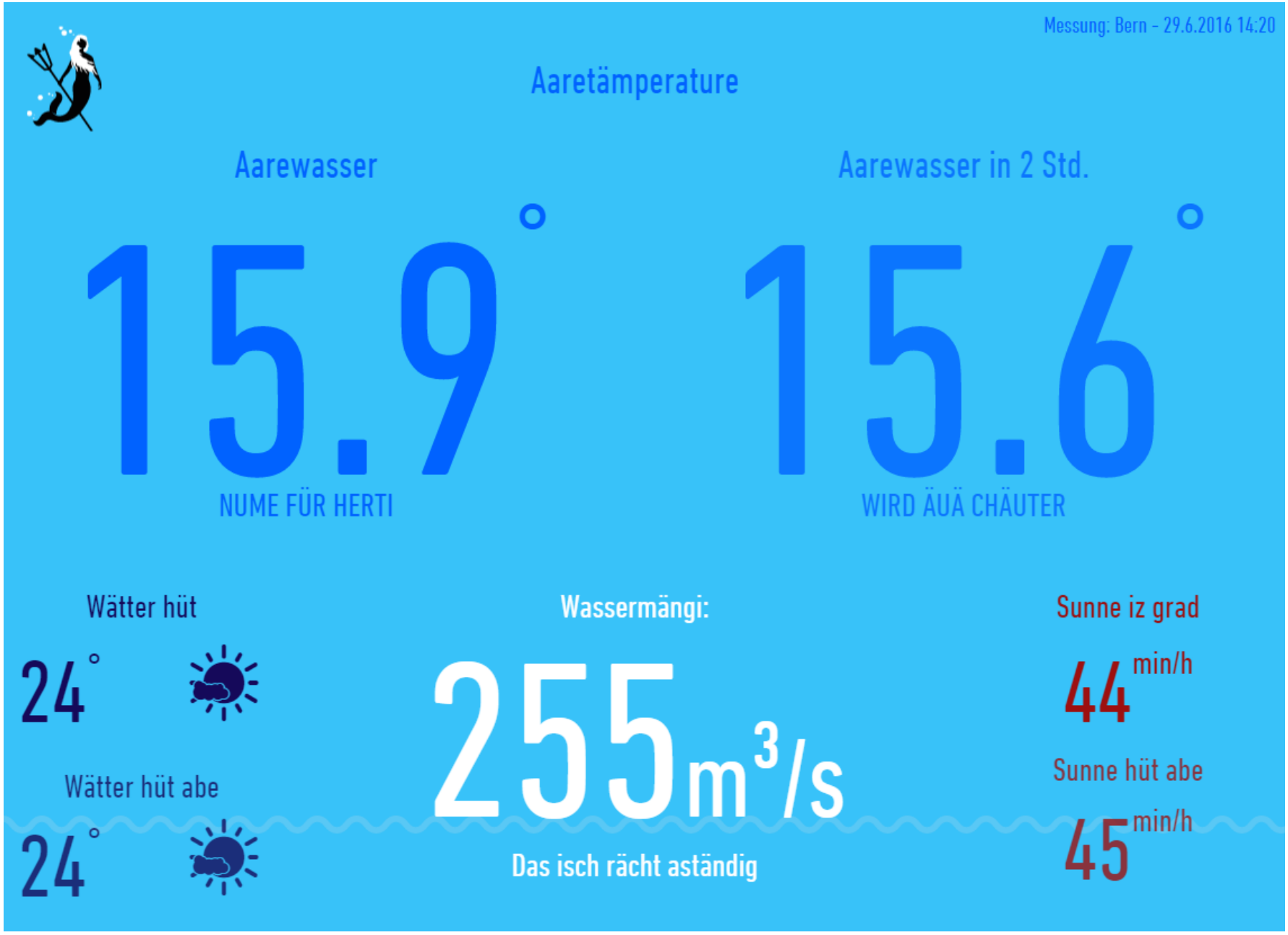
The interactions between these entities are creating new types of smart applications and services.

SENSORS + CONNECTIVITY + PEOPLE + PROCESSES

Making Sense of Data



Making Sense of Data



Smart Ecosystem

Sensoren

Big Data
Sammeln

Situative Relevanz
Auswerten

Smarte Services
Nutzen

Convenience Usability
Wirkung

Smart Ecosystem

SMART ECONOMY THINKING
am Beispiel Mobilität

HEUTE | ZUKUNFT

Nicht Auto, sondern **Mobilität**!

PRODUKT | WIRKUNG

Disrupt or be Disrupted

Früher

Kutsche

Faggoted
Grimm
Haussmann
Hureau
Passaglia
Rombach
Steyler
Staudner
...

Heute

Automobil

Audi
BMW
Mercedes-Benz
Toyota
Volkswagen
...

Zukunft

Eco-Mobility

Amazon
Apple
Google
Foxconn
Tesla
...

Smart City

Menschliches Nervensystem:

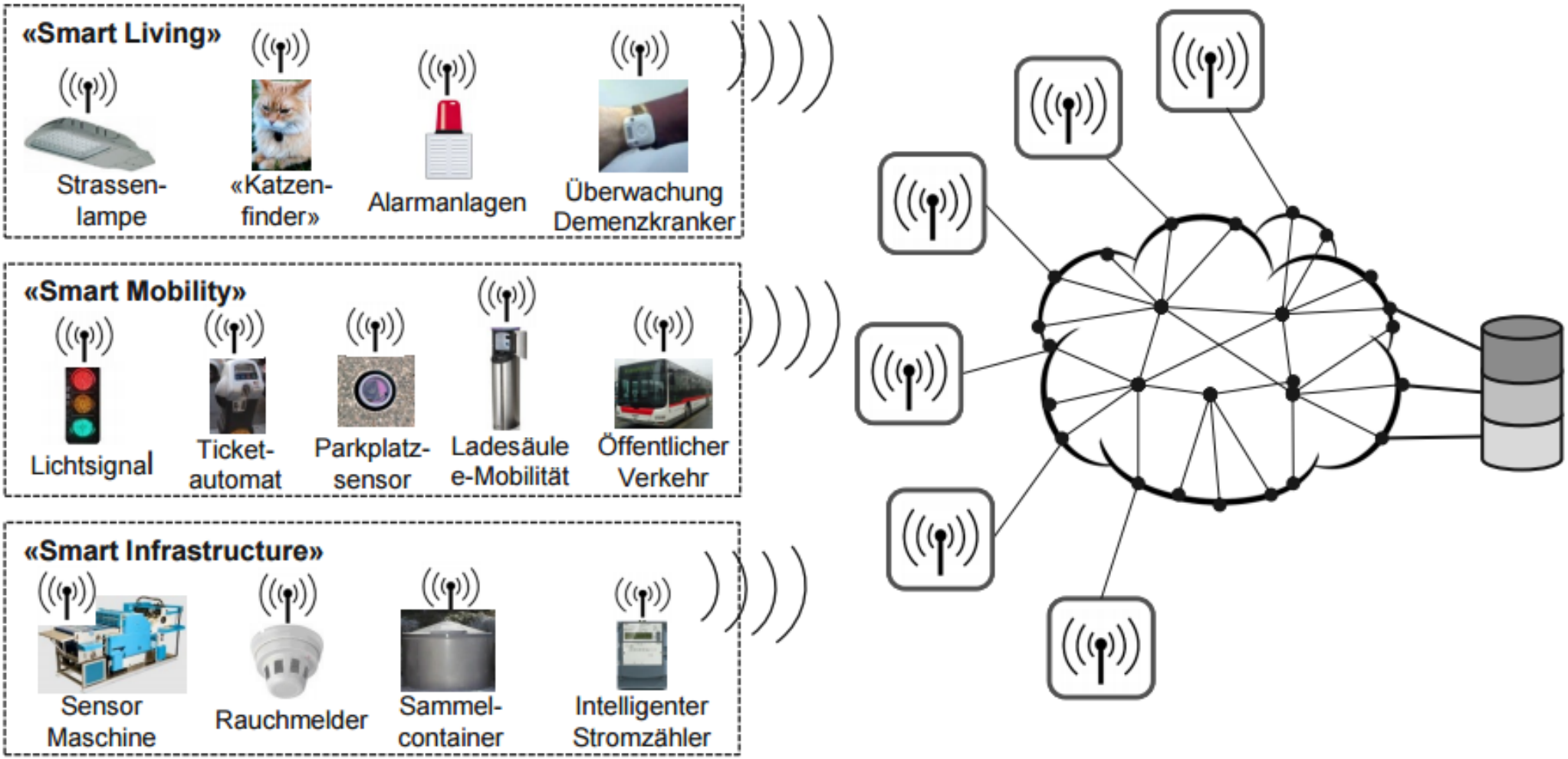
- Notwendig für gesamtheitliche Wahrnehmung
- Verarbeitung im Gehirn
- Basis für «intelligentes» Verhalten

Städtisches Netzwerk:

- Notwendig für gesamtheitliche Wahrnehmung
- Verarbeitung durch leistungsfähige Informatik unter Nutzung von «Big Data»
- Basis für «intelligentes» Verhalten

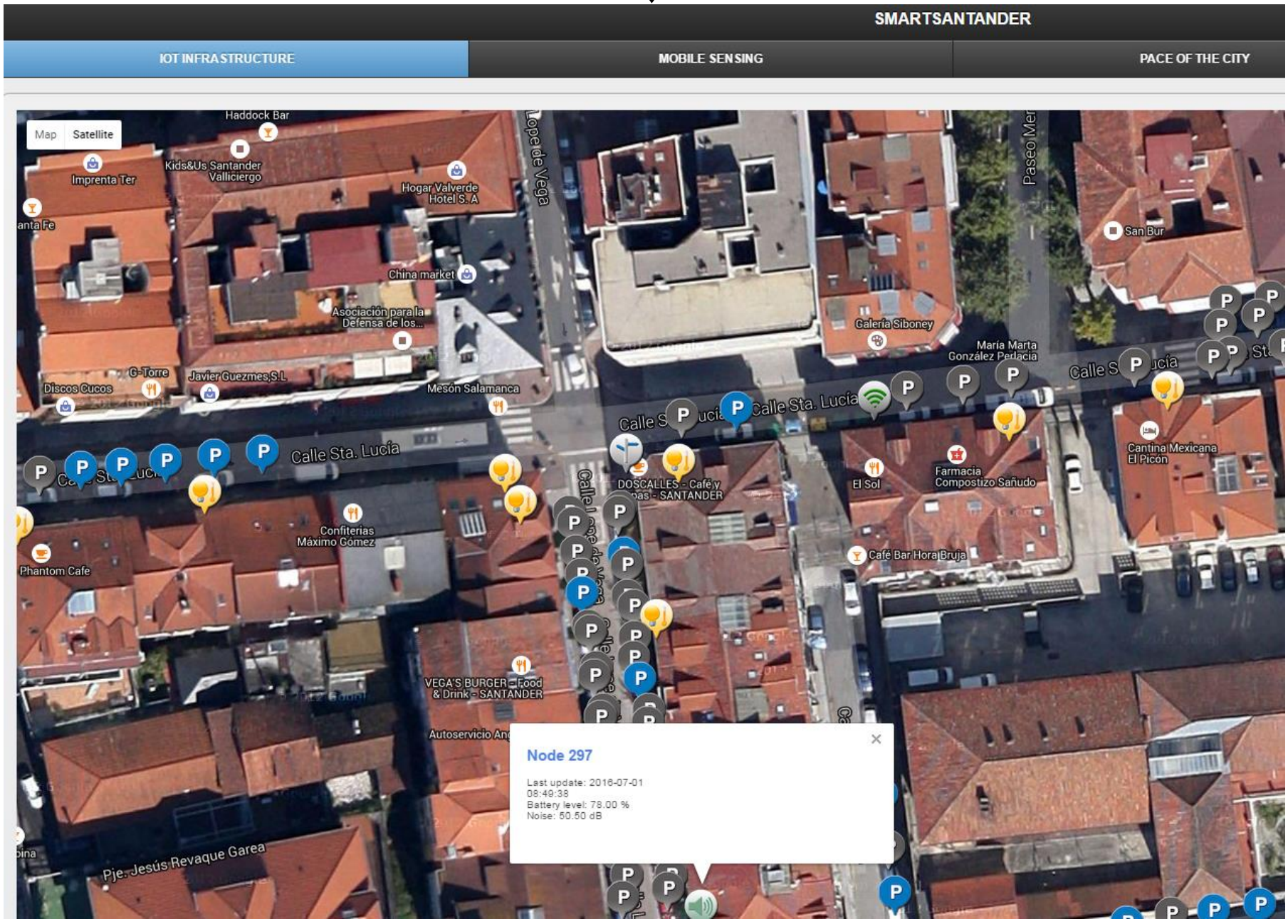
Analogie

Smart City



Objekte	Antennen Gateways	St.Galler Glasfasernetz	Daten-verarbeitung

Smart City

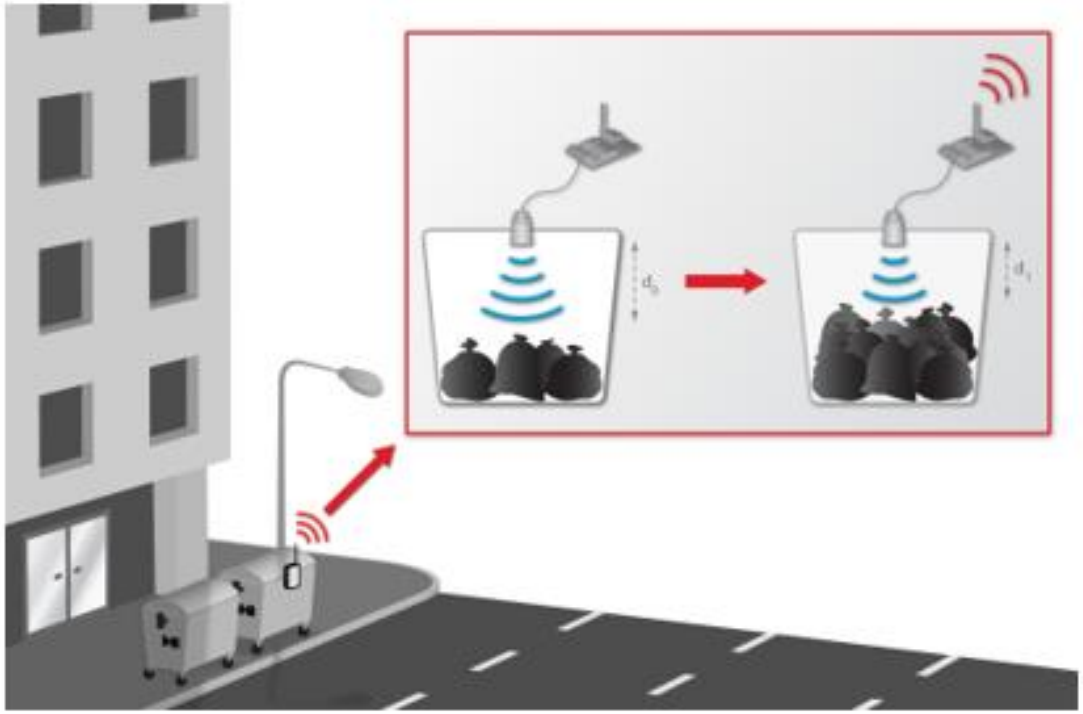

SMARTSANTANDER
IOT INFRASTRUCTURE
MOBILE SENSING
PACE OF THE CITY
Map
Satellite
Haddock Bar
Kids&Us Santander Valliciergo
Imprenta Ter
Hogar Valverde Hotel S. A
China market
Asociación para la Defensa de los...
Galería Siboney
María Marta González Perlacia
San Bur
G-Torre
Discos Cucos
Javier Guezmes,S.L
Mesón Salamanca
Calle Sta. Lucía
DOSCALLES - Café y tapas - SANTANDER
El Sol
Farmacia Compostizo Sañudo
Cantina Mexicana El Picón
Confiterías Máximo Gómez
Phantom Cafe
Café Bar Hora Bruja
VEGA'S BURGER - Food & Drink - SANTANDER
Pje. Jesús Revaque Garea
Node 297
Last update: 2016-07-01 08:49:38
Battery level: 78.00 %
Noise: 50.50 dB

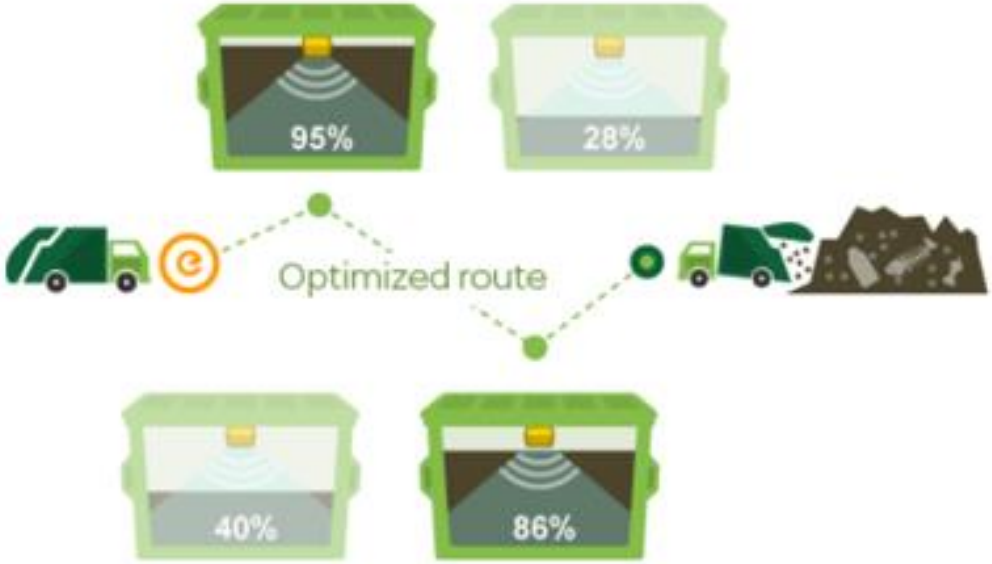
Smart City

Wireless Transceiver
Photocell
App Based Control
Concealed Placement Speaker
RGBA Notification
Image Sensor / Pedestrian Counter
MAIN ST
Digital Street Sign
Digital Signage
Façade Lighting
Push To Talk System
Seismic Sensors
Water Detection

libelium
LoRa

Smart City





Smart City

Smart Building

Deloitte.

SMART EDGE

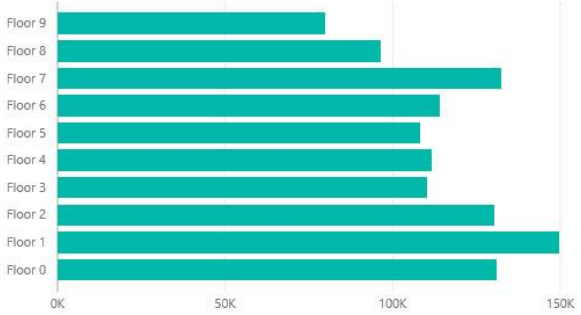
Actual visitors

237

Registered visitors

2224

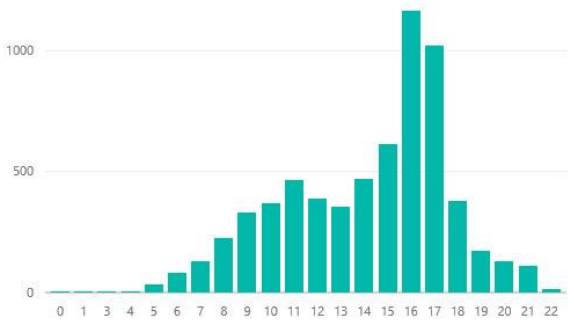
Average energy use (kW)
BY FLOOR



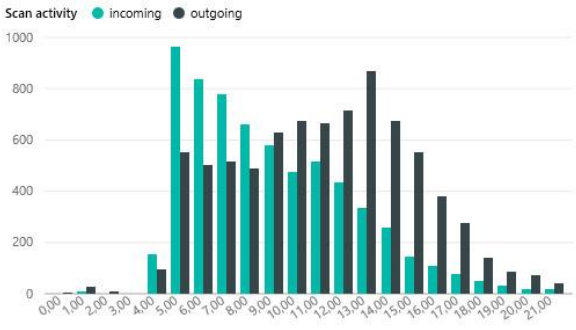
Average building temperature

19,48

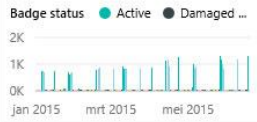
Building visitors
BY HOUR



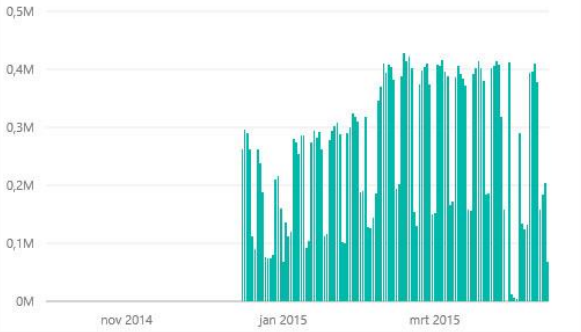
Visitors (Lenel)
PER HOUR OF DAY



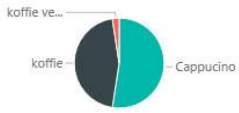
Badge scans per badge status
DOORDAY, BADGE STATUS



Average energy consumption (kW)
BY DATE

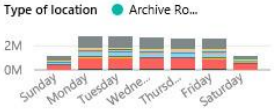


Coffee Use - example
BY PRODUCT

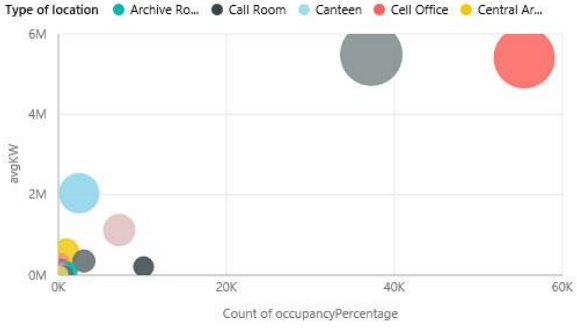


Current availability Deloitte parkin...

Average energy consumption (kW...
BY TYPE OF LOCATION

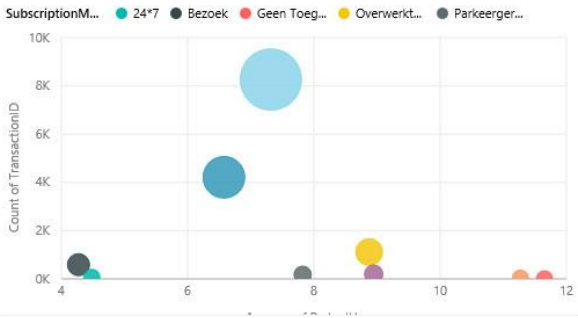


Energy consumption and room occupancy
BY TYPE OF LOCATION

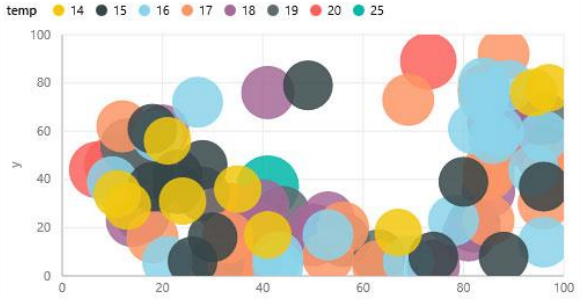


Registered Guest (Axxerion)
PER START TIME

Number of parking visitors and hours parked
BY SUBSCRIPTIONMODEL

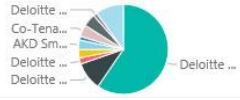


Temperature - example
ON FLOORPLAN



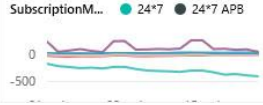
Visitors with Car

Badge Types



Smartcard Swipes
PER DAY

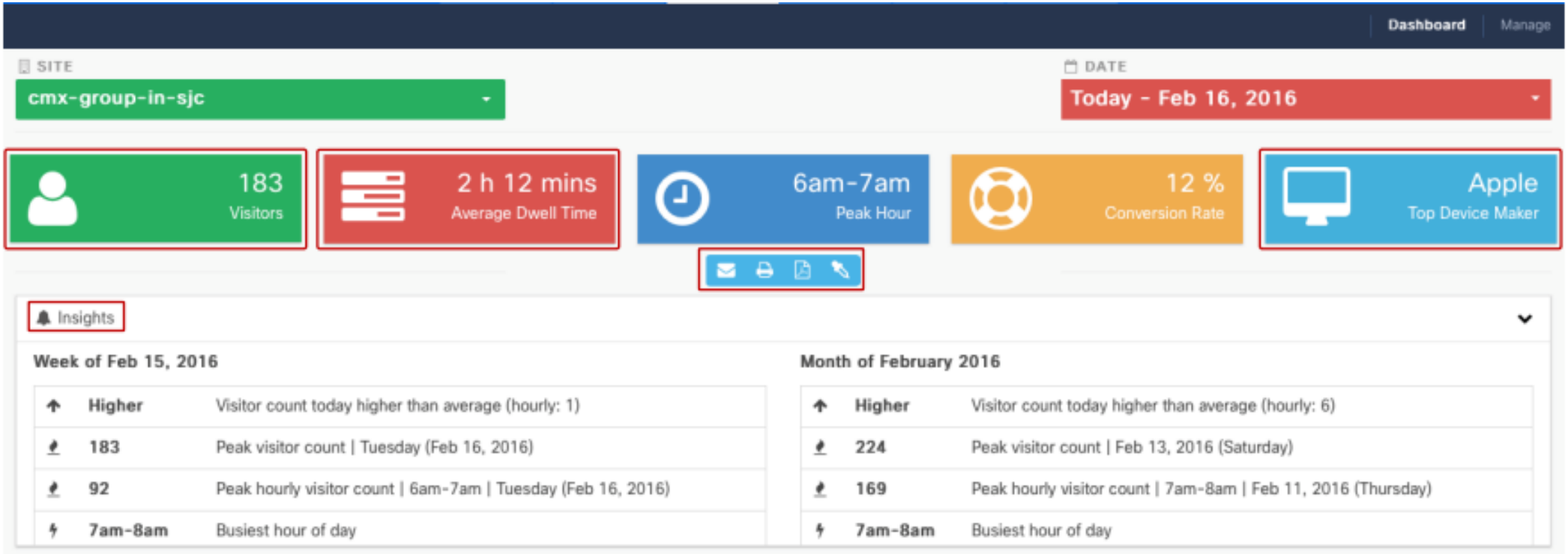
Free Parking Pplaces



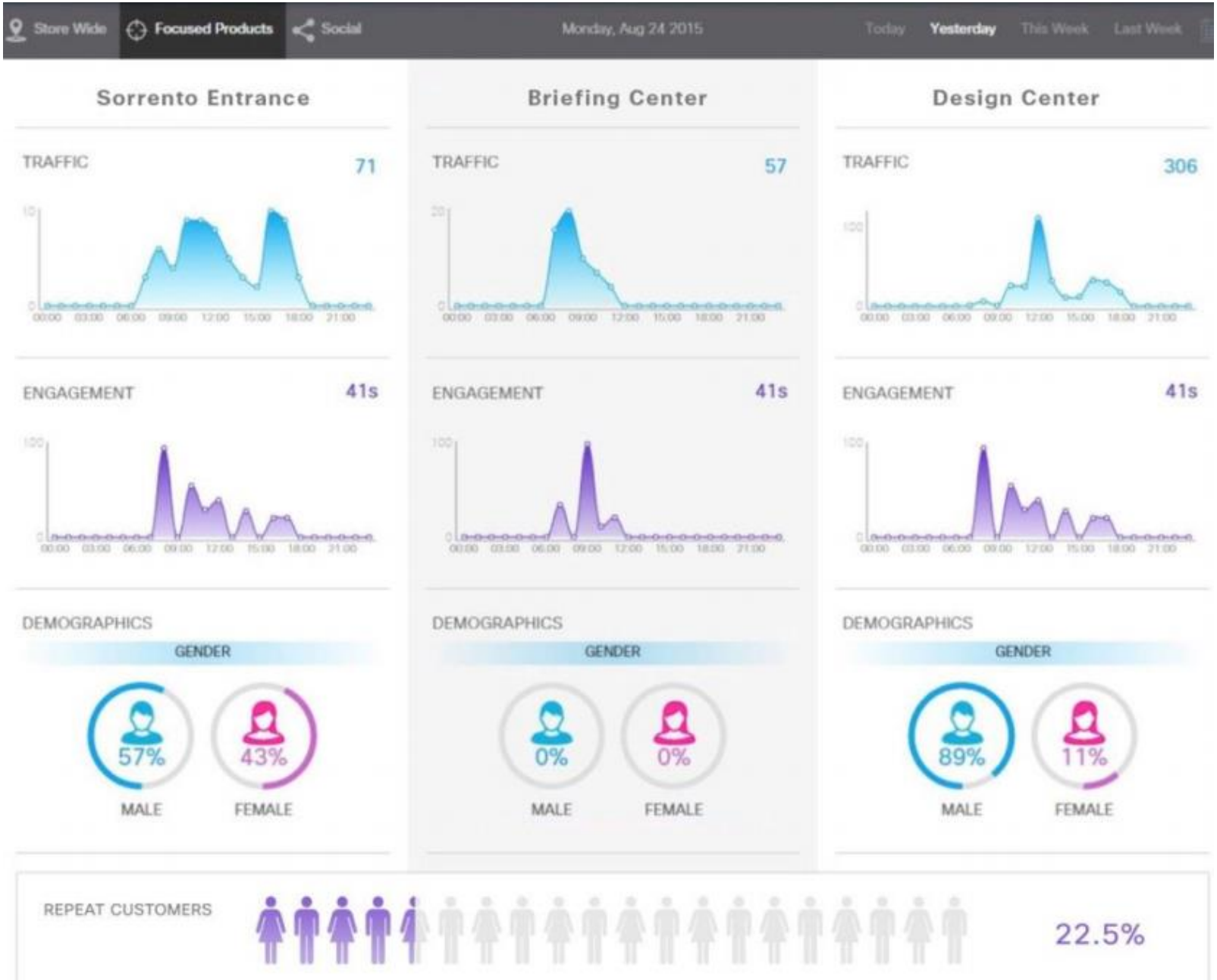
Retail 360°

Kunden, die
dieses Produkt
gekauft haben
kauften
auch
DON MARIO
DON MARIO
© romankmenta.com

Retail Analytics

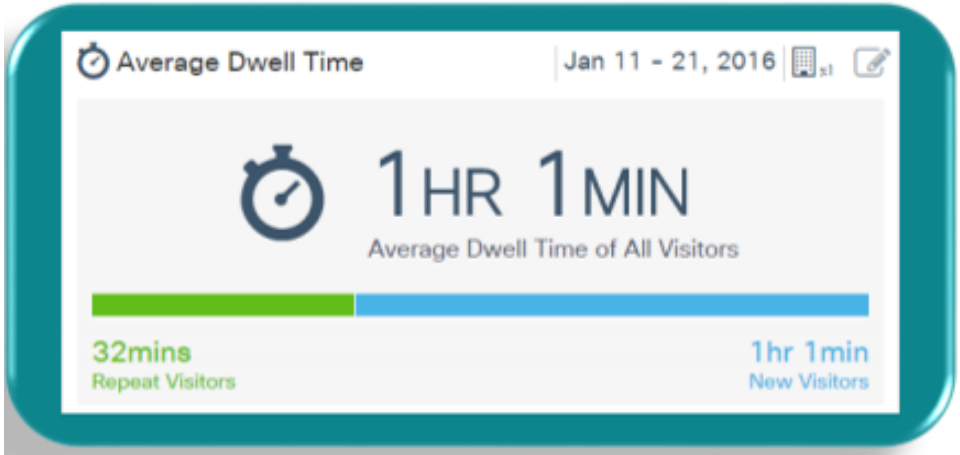


Retail Analytics



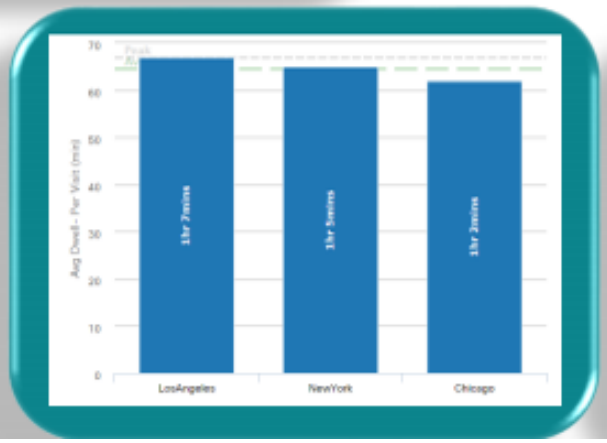
Retail Analytics

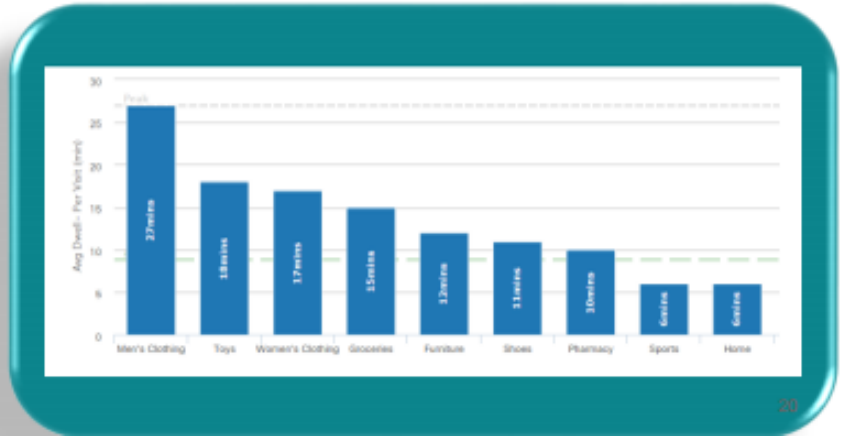
How long do customers spend in my store?



In different departments?

In different stores?





Retail Analytics

Which departments do people spend time in?

Expected: most people walk through the entrance area

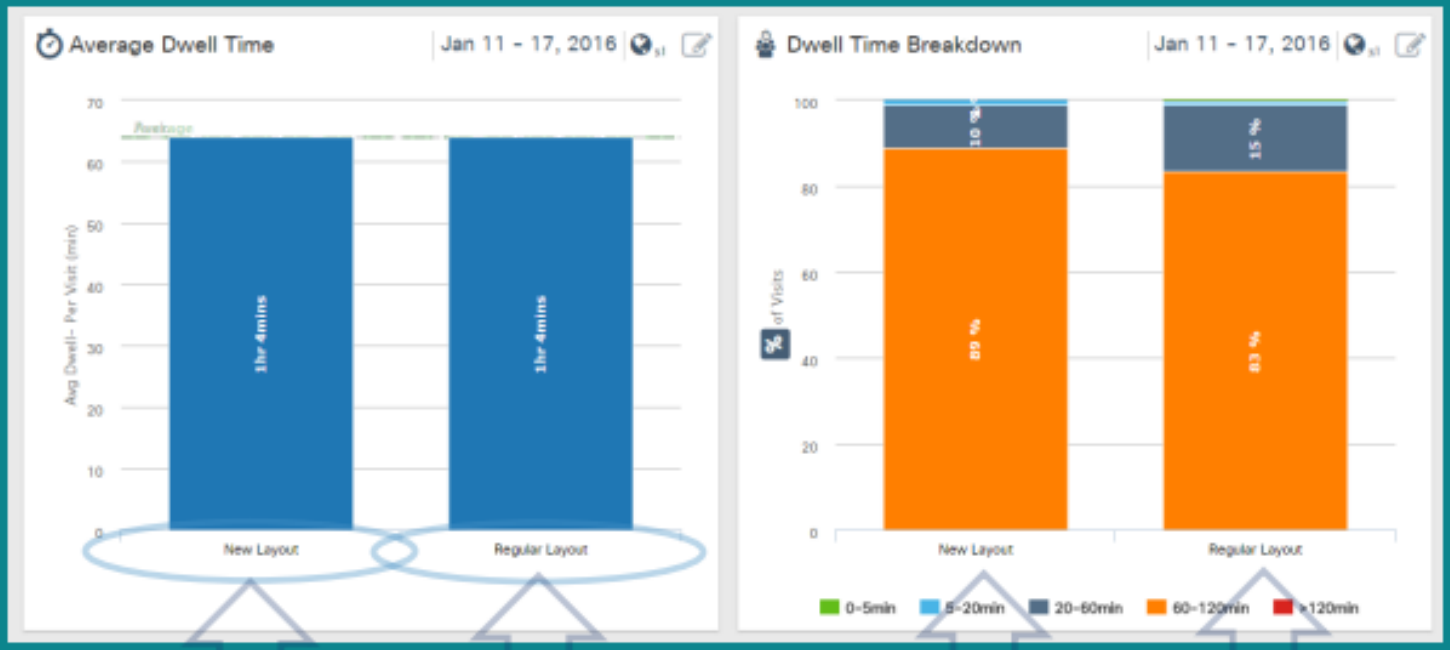
Could be because of the video game booths

Why are shoppers not spending more than 20 minutes?

Coffee is mostly grab and go

Retail Analytics

AB testing – is the new store layout effective?



Stores where we are testing the new layout

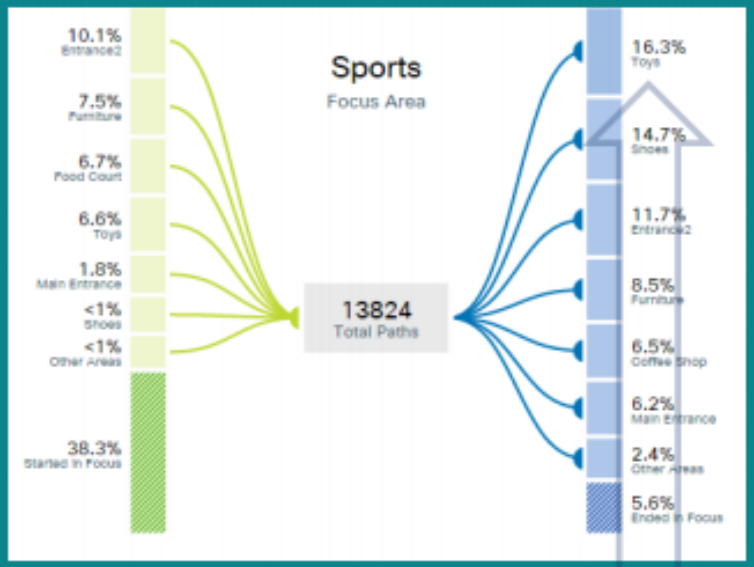
Stores that are still on the regular store layout

A much higher percentage (89%) of visitors to stores with the new layout spend more than an hour in the store!

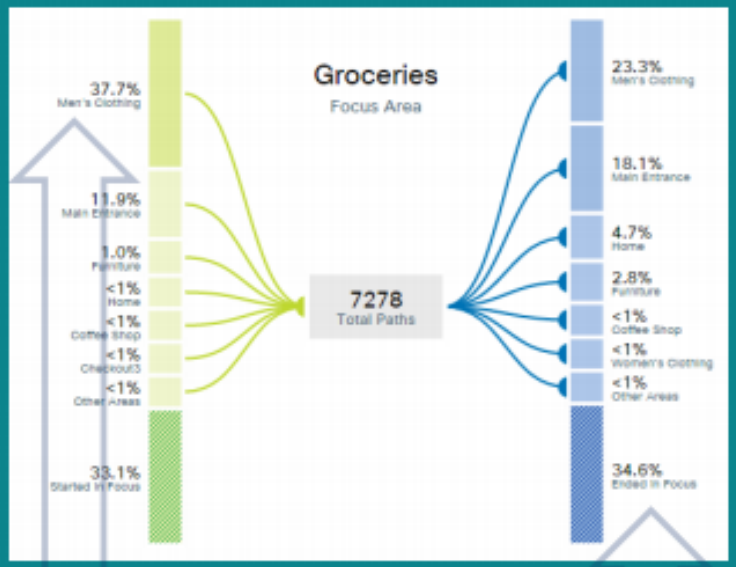
Retail Analytics

What are the paths visitors take in the stores?

Where do people go after visiting a department?



16.3% go to the toy department after the sports department



Many visitors stop at the men's clothing department before going to the groceries department

More than a third of visitors go home after visiting the Groceries department

Retail Analytics

Kool Zone Barrelytics

Beer Barrel Tracking

BECK'S

Beer Remaining:

20%

Replace

WARSTEINER

Beer Remaining:

35%

Low

HEINEKEN

Beer Remaining:

50%

OK

BERLINER KINDEL

Beer Remaining:

100%

Full

Ambient Temperatures

3 °C	COOLANT TEMP
10 °C	STORAGE TEMP
25 °C	OUTSIDE TEMP

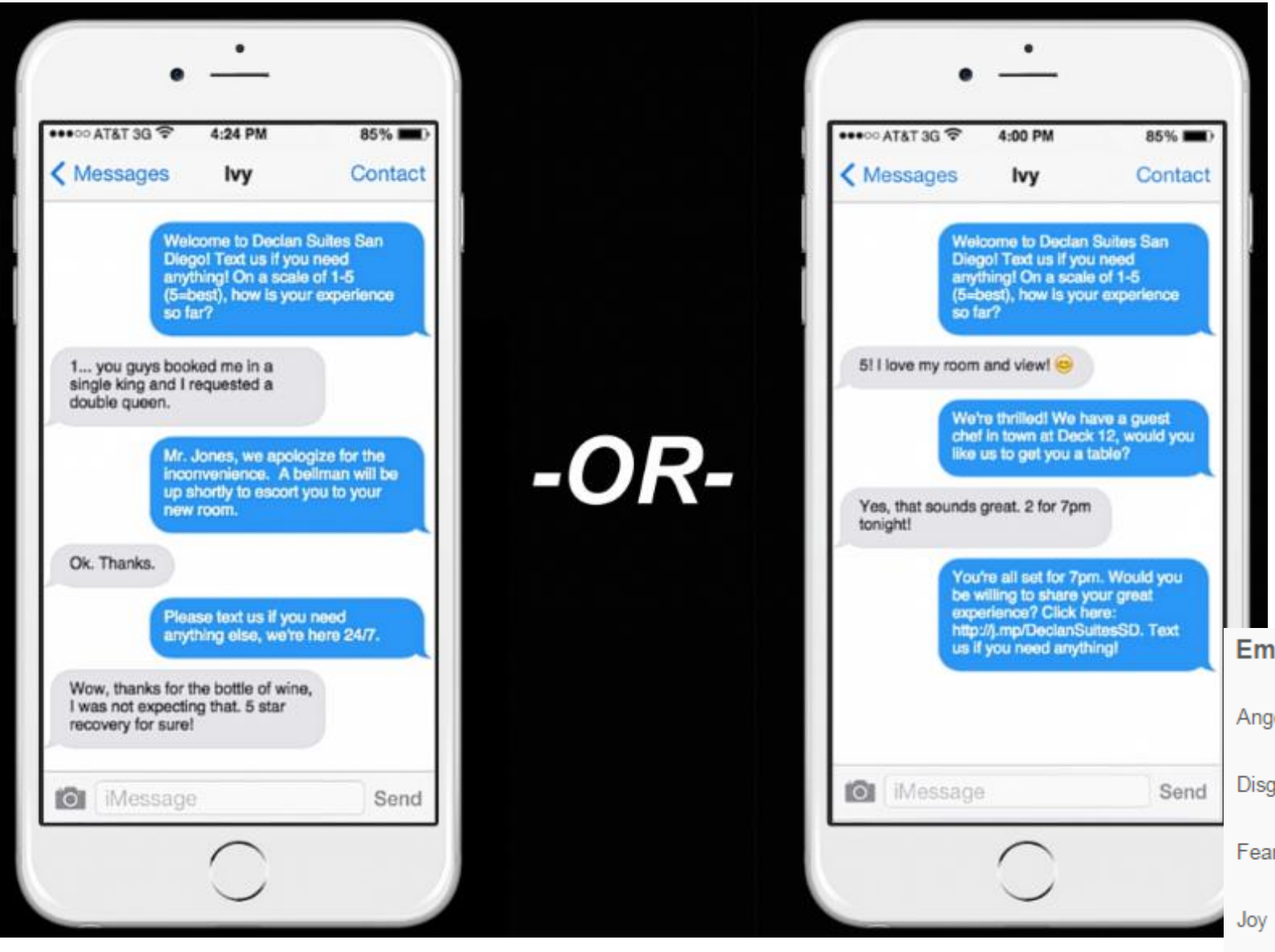
Revenue Loss

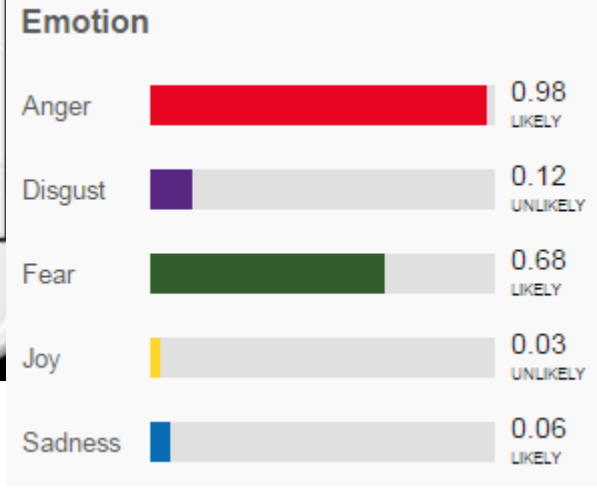
850 L	SALES
940 L	ACTUAL
90 L	WASTAGE

Elevator

Washing Machine

IBM Watson


AT&T 3G 4:24 PM 85%
Messages Ivy Contact
Welcome to Declan Suites San Diego! Text us if you need anything! On a scale of 1-5 (5=best), how is your experience so far?
1... you guys booked me in a single king and I requested a double queen.
Mr. Jones, we apologize for the inconvenience. A bellman will be up shortly to escort you to your new room.
Ok. Thanks.
Please text us if you need anything else, we're here 24/7.
Wow, thanks for the bottle of wine, I was not expecting that. 5 star recovery for sure!
iMessage Send
-OR-
AT&T 3G 4:00 PM 85%
Messages Ivy Contact
Welcome to Declan Suites San Diego! Text us if you need anything! On a scale of 1-5 (5=best), how is your experience so far?
5! I love my room and view!
We're thrilled! We have a guest chef in town at Deck 12, would you like us to get you a table?
Yes, that sounds great. 2 for 7pm tonight!
You're all set for 7pm. Would you be willing to share your great experience? Click here: http://j.mp/DeclanSuitesSD. Text us if you need anything!
iMessage Send


Emotion
Anger 0.98 LIKELY
Disgust 0.12 UNLIKELY
Fear 0.68 LIKELY
Joy 0.03 UNLIKELY
Sadness 0.06 LIKELY

Gemeinsamkeiten

Verteiltes **Sensornetzwerk**

Verbunden mit **Internet**

Aggregation von Daten

Offenlegen von **Antworten**

Wertschöpfung durch **Handeln**

SPOUD

Transforming Data into Action

TECHNOLOGIE

Architektur IoT Lösung

Sensoren

Real-Time Analytics

Actions

Challenges

Datenvolumen

Datenkonsistenz

Sicherheit und Vertraulichkeit

Daten-Magie

Qualität

Reproduzierbarkeit

Infrastruktur

Datenvolumen

Millionen von **Sensoren**

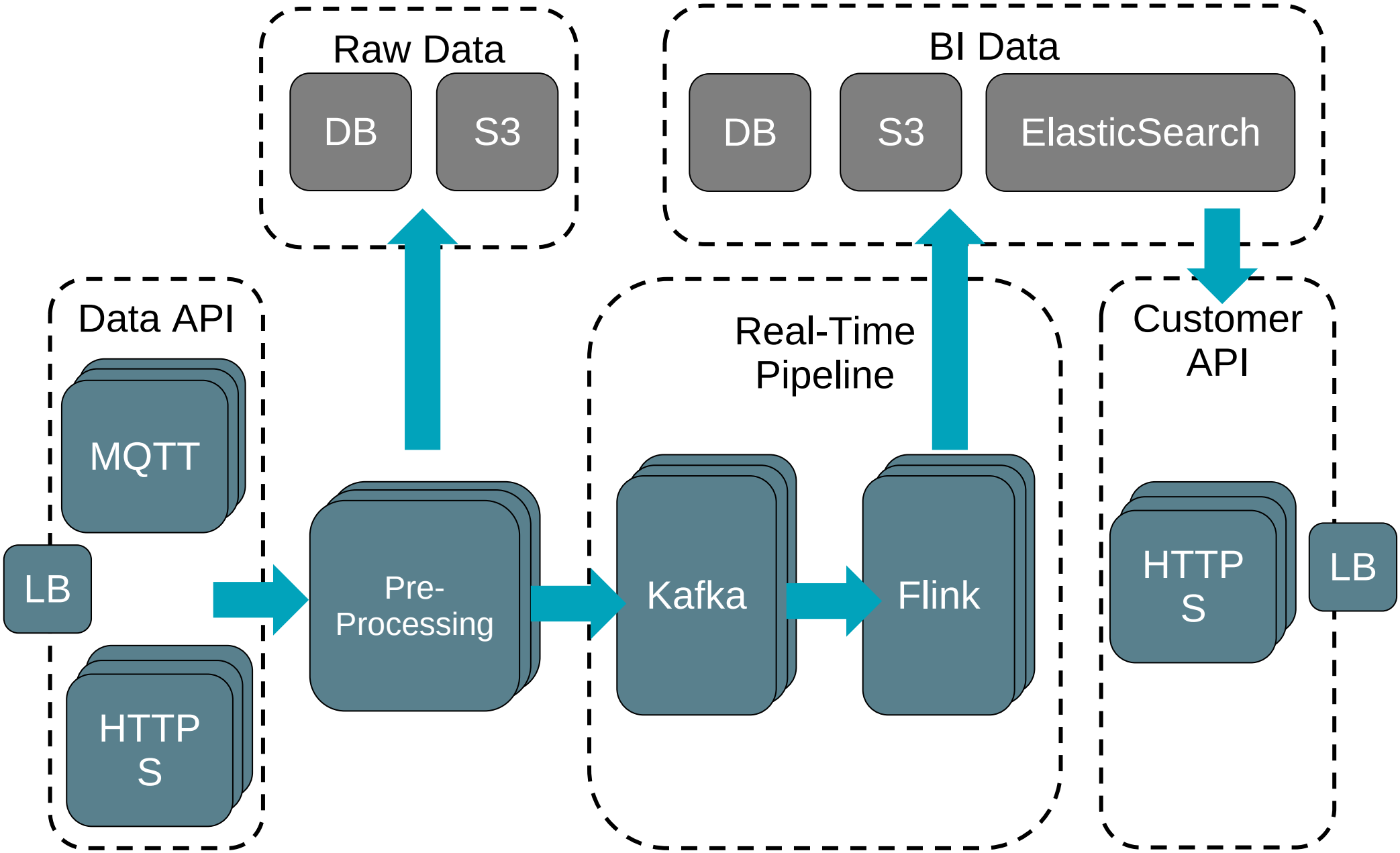
Millionen von **Events** pro Sekunde

Speicherung rohe Daten und Analysen

Echtzeit Anspruch

Stateful Algorithms

Platform Architecture



Sicherheit und Vertraulichkeit

Daten sind das **Kapital** der IoT

Sicherheit und **Vertraulichkeit** sind zentral

Verschlüsselung von Daten

Anonymisierung/Reduktion direkt auf Sensor

Analytics

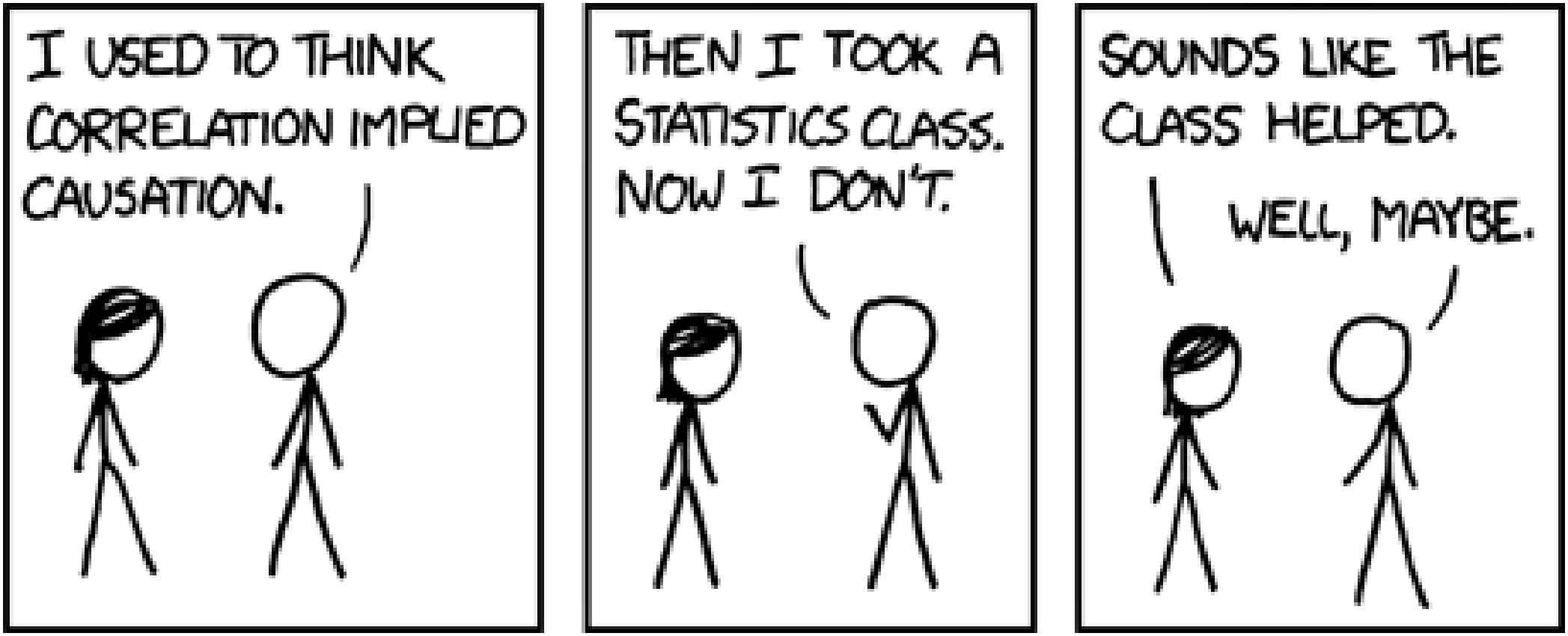
Verschiedene Wege führen nach Rom

Algorithmus **Tuning** sehr aufwendig

Systematische **Verifikation** schwer

Korrelation nicht **Kausalität**

Analytics


I USED TO THINK CORRELATION IMPLIED CAUSATION.
THEN I TOOK A STATISTICS CLASS. NOW I DON'T.
SOUNDS LIKE THE CLASS HELPED.
WELL, MAYBE.

Qualität

Komplexes **System**, komplexe **Algorithmen**

Blackbox Integrationtesting mit **Referenzdaten**

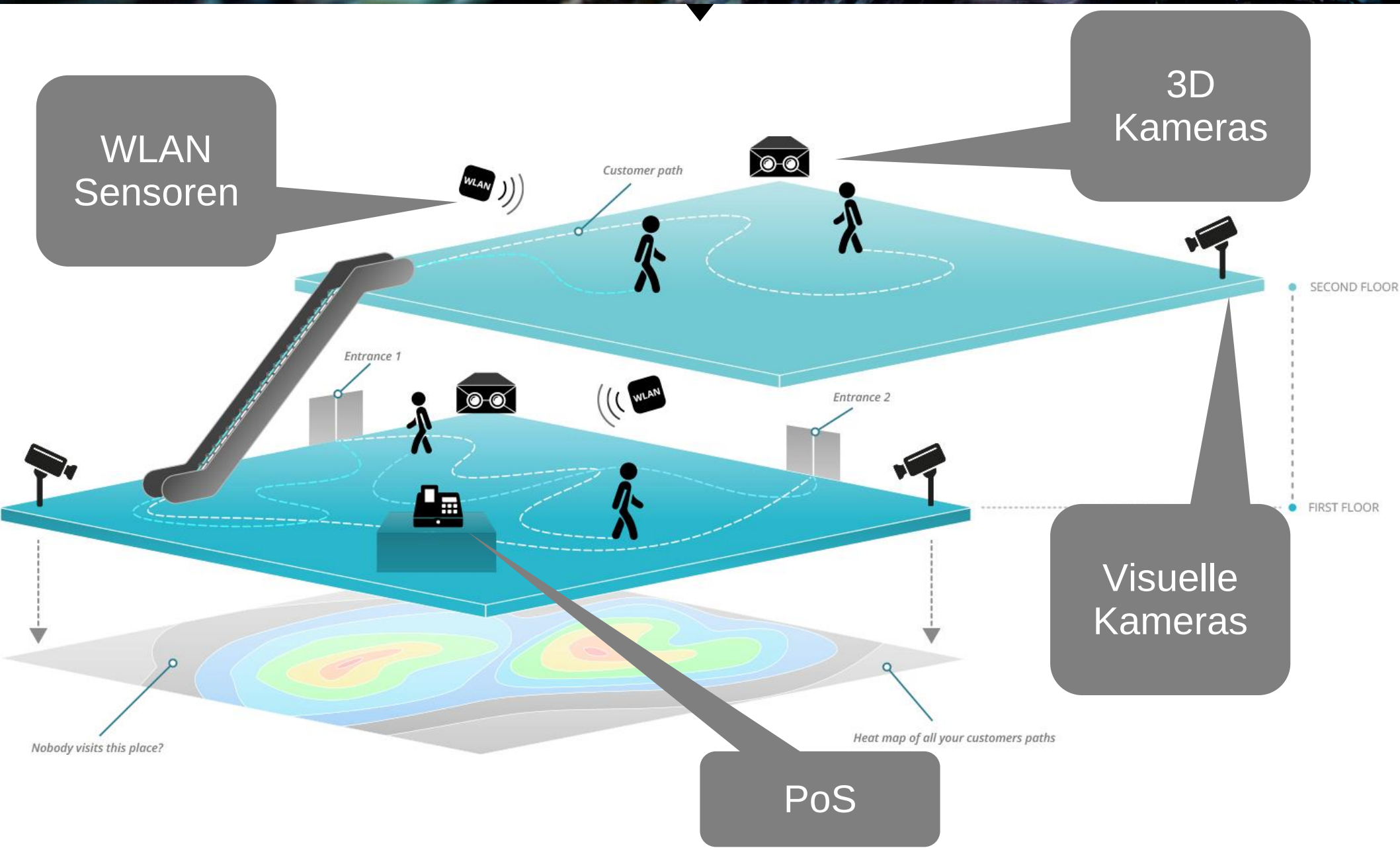
Boundary Check zur Plausibilitätsprüfung

Anomaly Detection auf Lösungsmenge

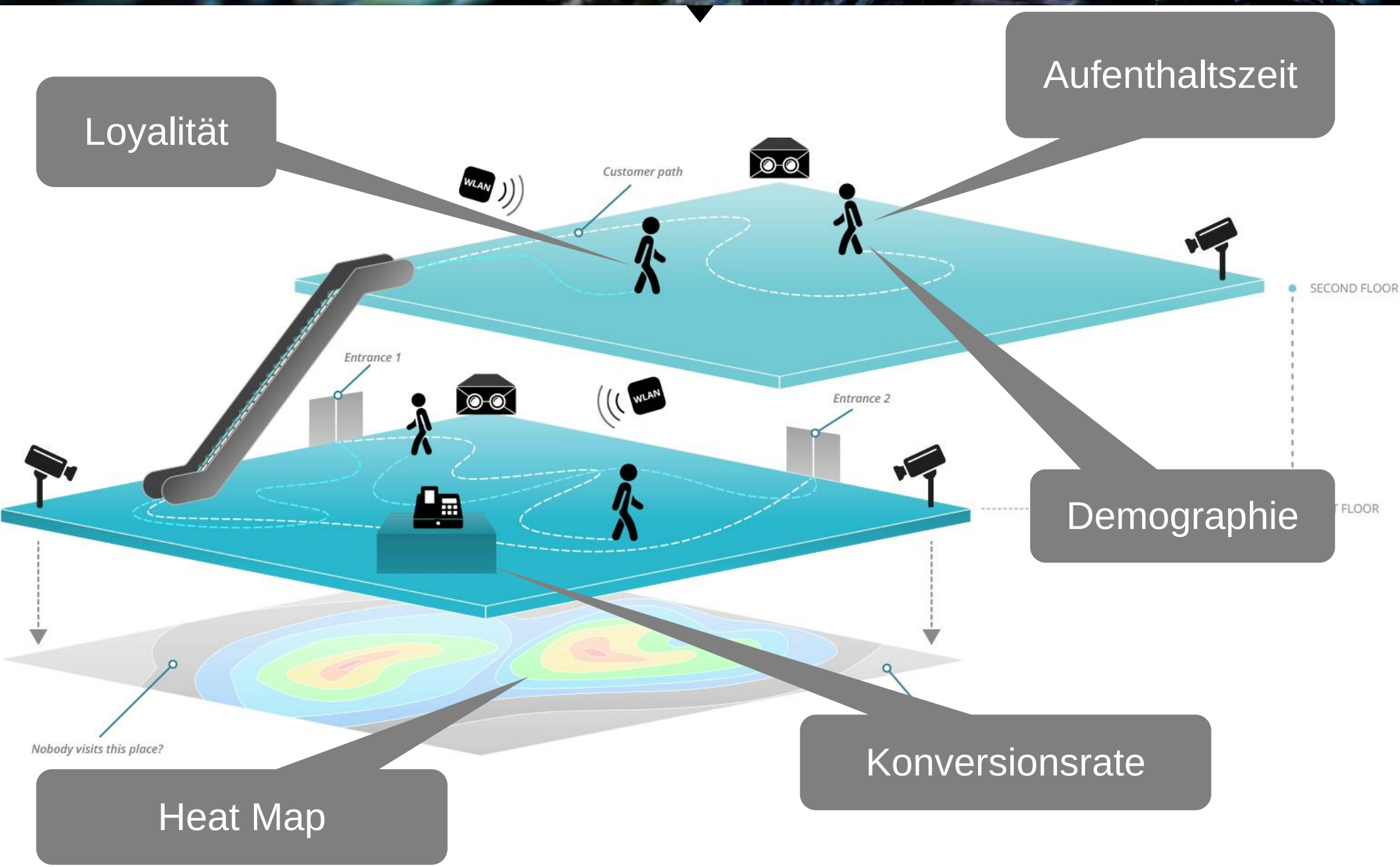
ERFAHRUNGEN

Retail Analytics

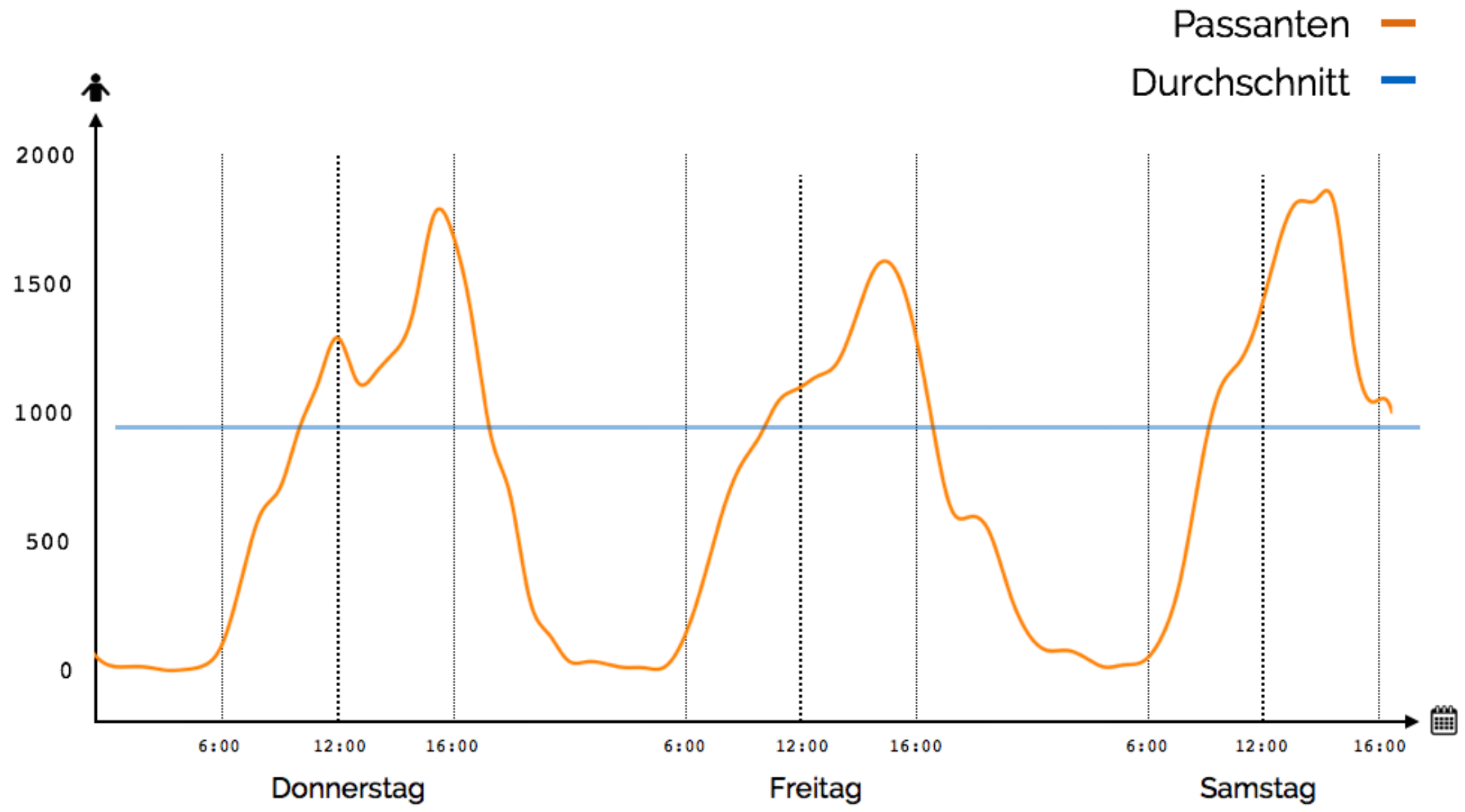
storalytics



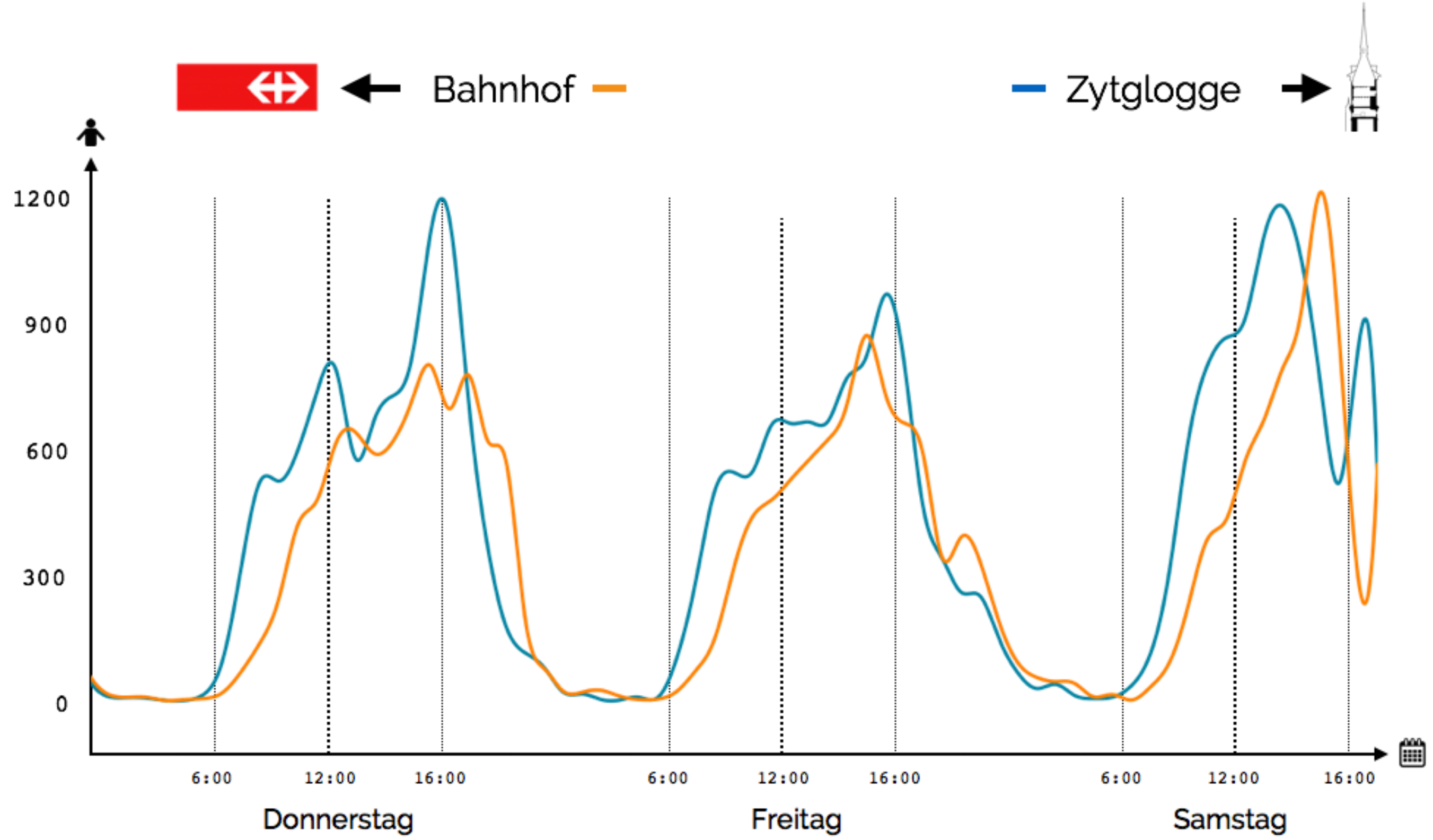
storalytics



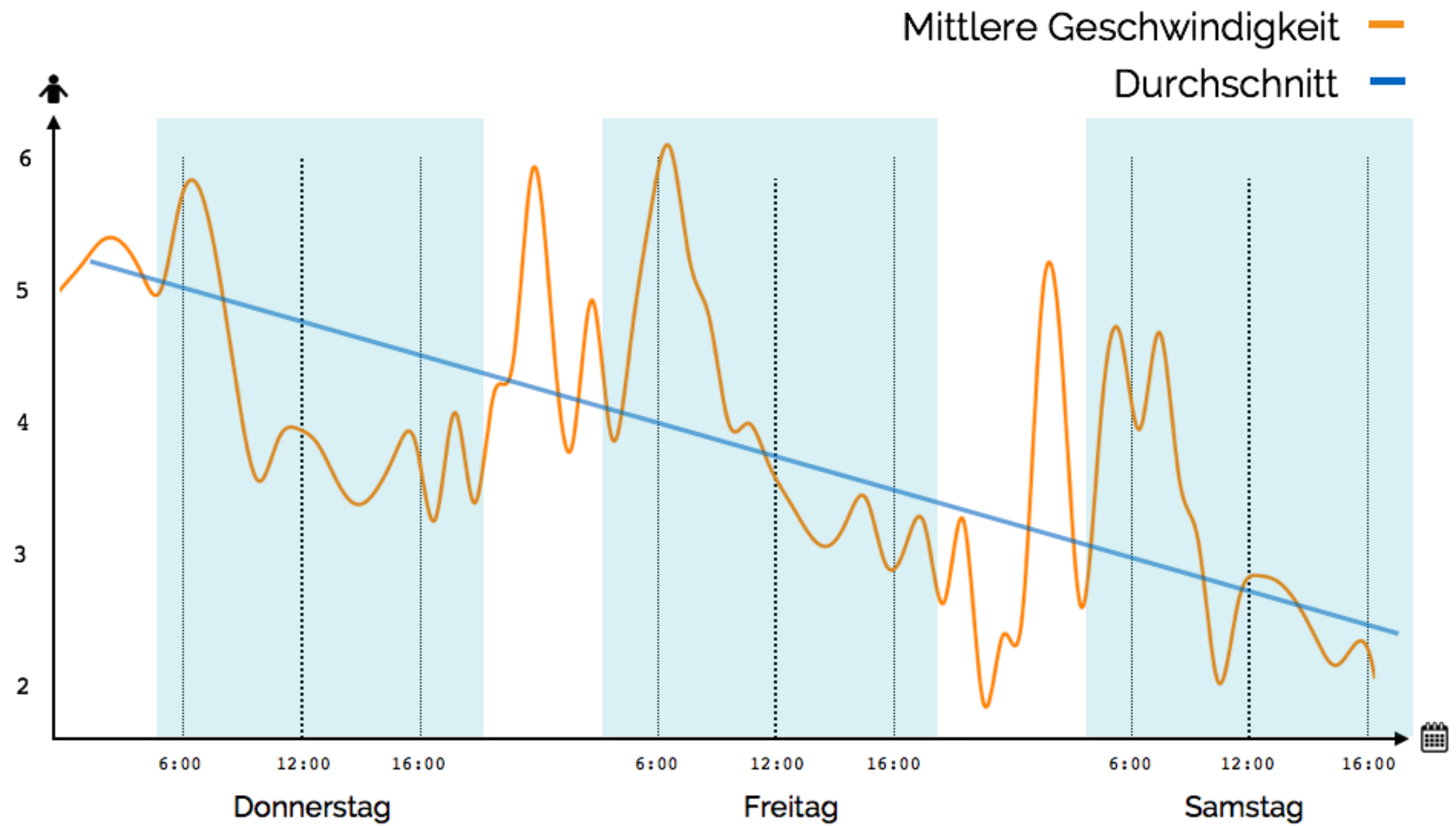
Passanten pro Stunde



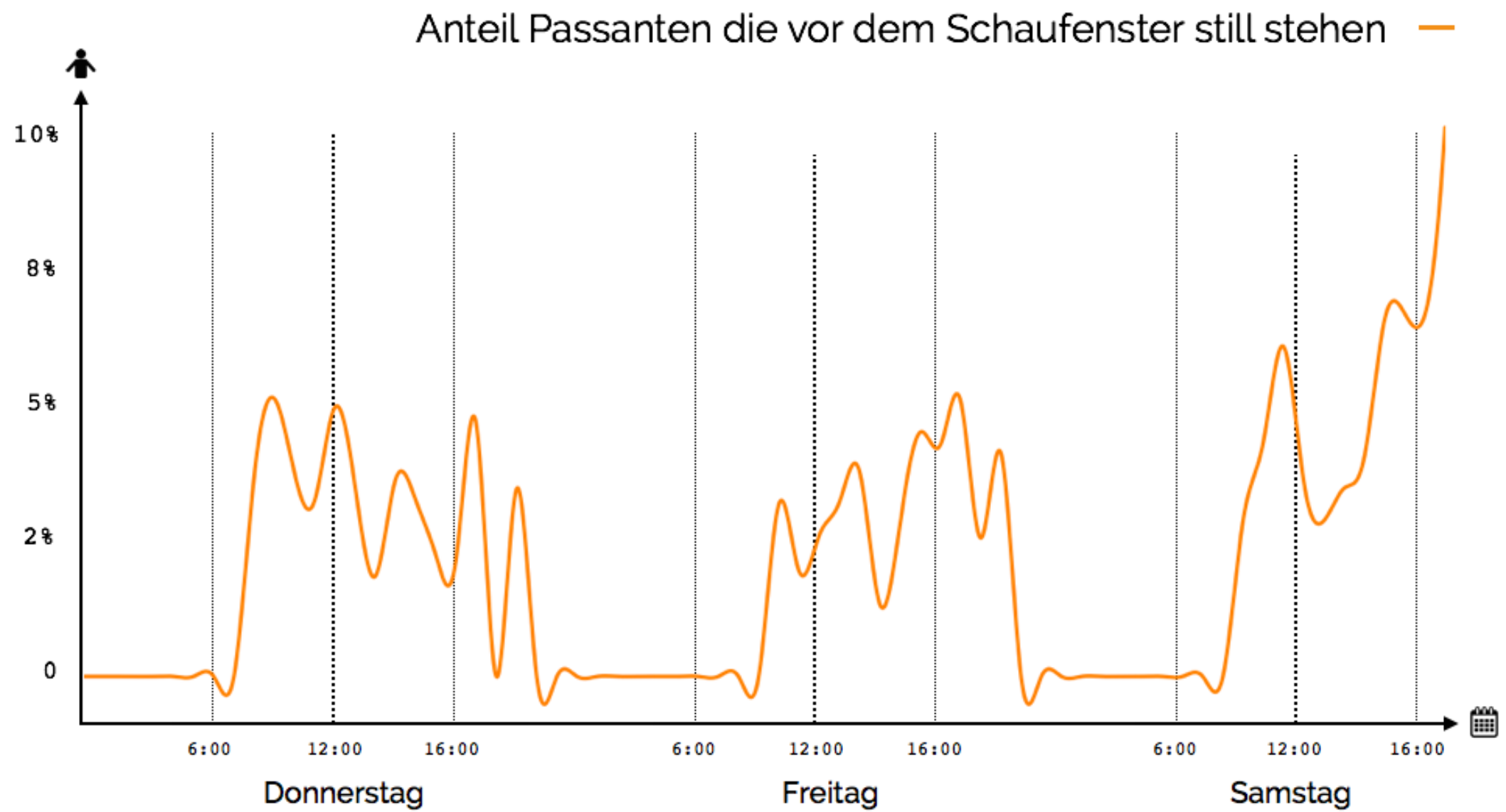
Passanten nach Richtung



Passanten Geschwindigkeit



Passanten die Stillstehen



Erfahrungen aus Retail

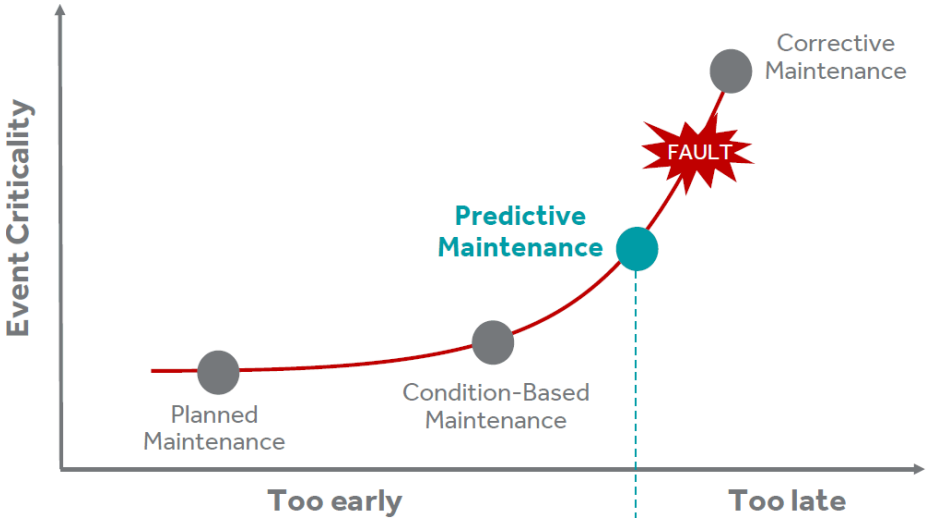
Qualitätssicherung nicht trivial

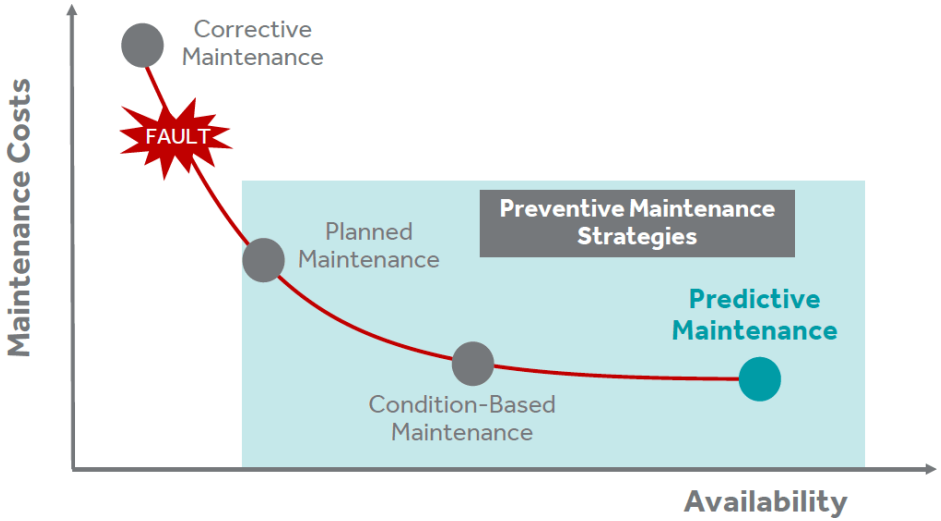
Sensorik grösste Fehlerquelle

Disruptive Technologie

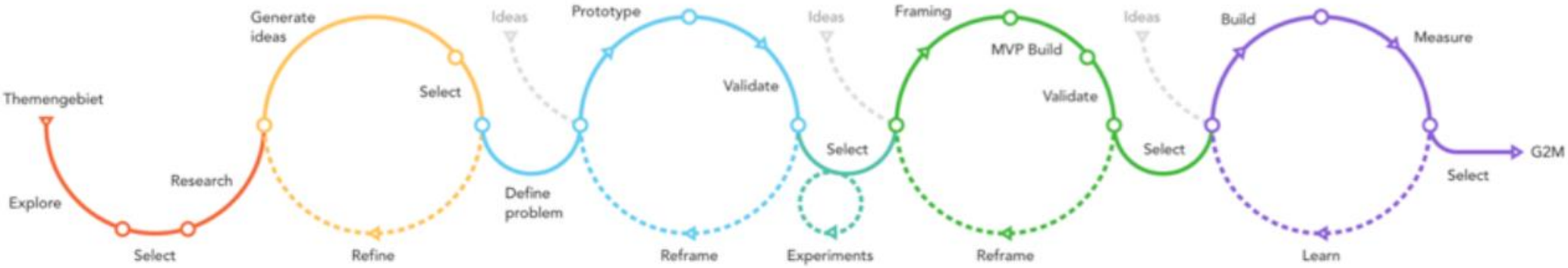
ALPIQ – AUDIO FINGERPRINT

Eine Vielzahl von Industrie Kälteanlagen bei Kunden
Kontinuierliche Service Checks Vor-Ort erforderlich
Probleme frühzeitig erkennen
Zur richtigen Zeit am richtigen Ort





ALPIQ – AUDIO FINGERPRINT



ALPIQ – AUDIO FINGERPRINT

System Monitoring NEEO

System Monitoring NEEO

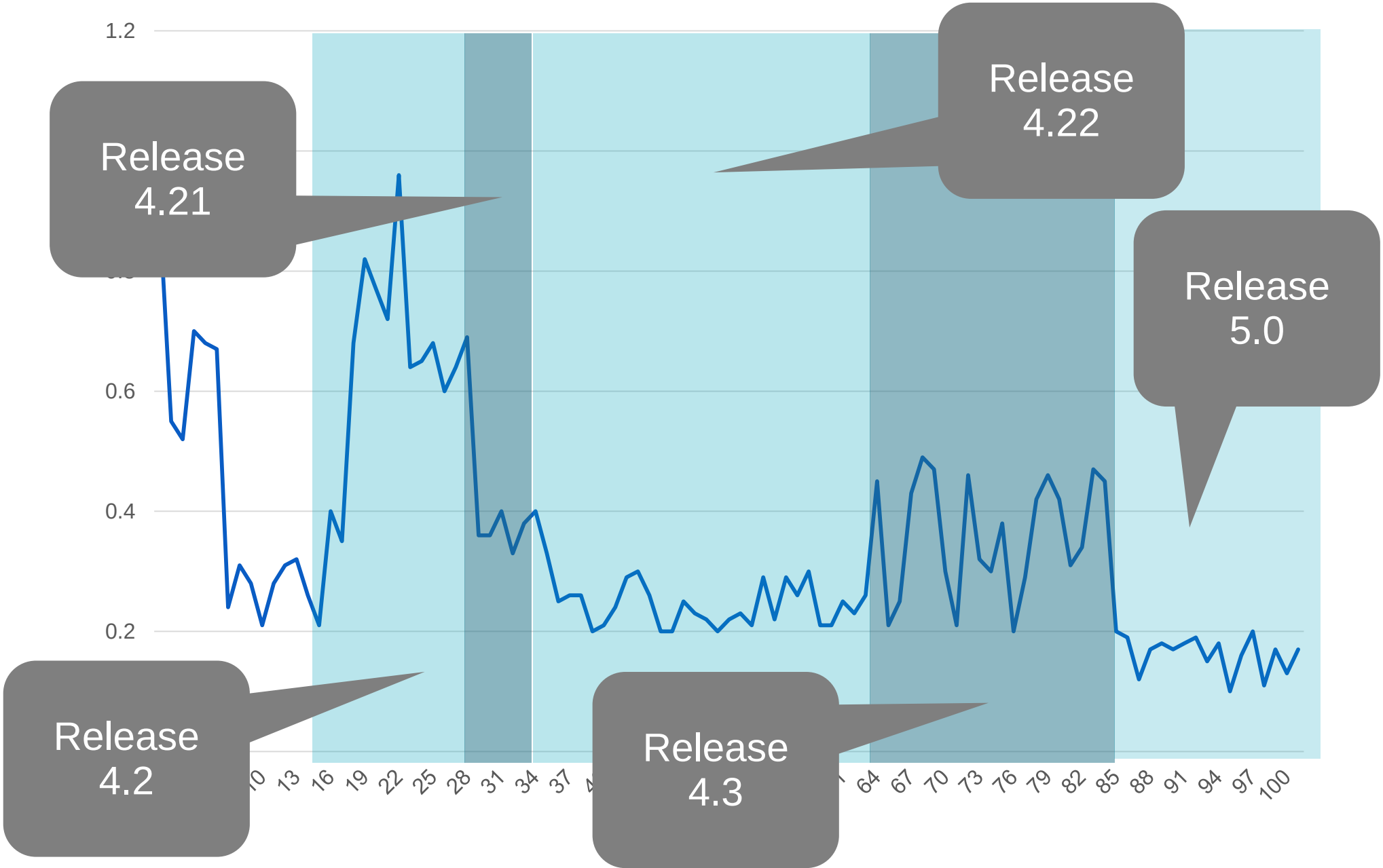
NEEO ist ein **Home Automation** Hub

Erfolgreichstes Automation Projekt Kickstarter

Testing Phase mit 100 Betatester

Monitoring **Produktqualität**

Produktqualität Index



ZU GUTER LETZT

Answers need Questions

Entscheidend: **Richtige Fragen** stellen

Interdisziplinäres Team

Fachlichkeit, Wirtschaftlichkeit und Technologie

Inkrementelles Vorgehen

Disruptiv

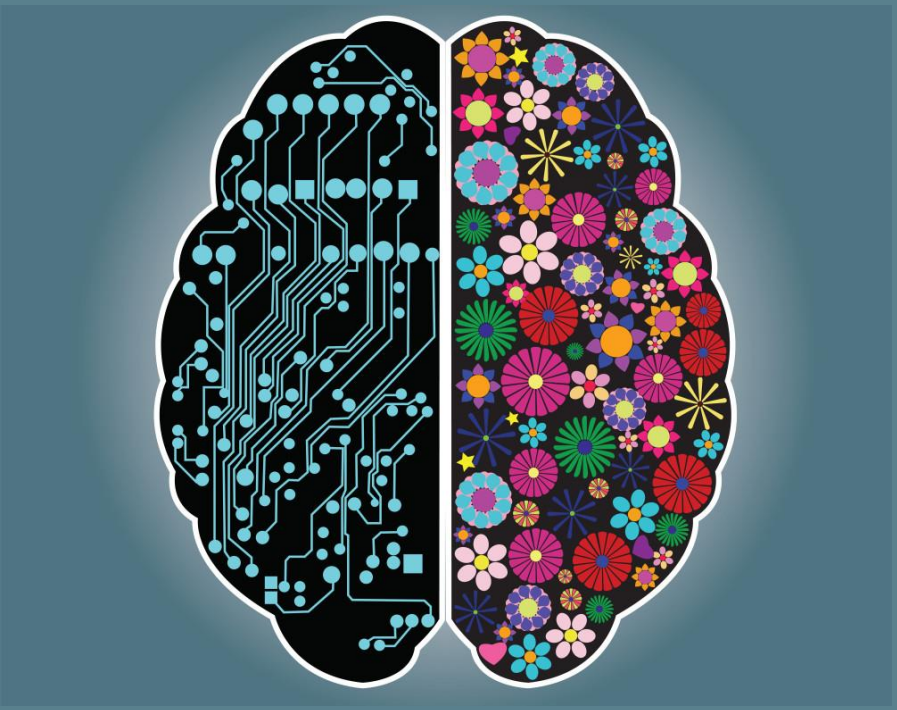
Datenbasierte Entscheidungsfindung ist real

Real-time erlaubt zeitnahe Entscheidungen

Erlaubt Änderungen der **Geschäftsprozesse**

Bedingt Änderungen im **Mindset**

Decision Making Sweet Spot



Schulung und **Validierung**
der eigenen **Intuition**

Gegen **Vorurteile schützen**

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